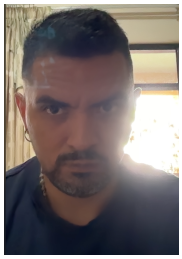
Omar Corona Tejeda



AI TRAINER & EVALUATOR | MATHEMATICAL REASONING & FORMAL VERIFICATION

Mexico City, Mexico | (+52) 272-283-9135
omarct1989@gmail.com | omarct1989@ciencias.unam.mx
LinkedIn | Mathematical Physics Project

Mathematician and physicist with experience in AI training, response evaluation, and university-level teaching. Develops advanced mathematical problems, rubrics, and reference solutions, with formal verification in Lean 4 and Mathlib. Combines analytical reasoning, multimodal data quality assurance, and clear technical communication.

CORE EXPERTISE

Mathematical Reasoning & Formal Verification: Advanced problem design, proof analysis, Lean 4 & Mathlib.

AI Training & Evaluation: Response assessment, rubrics, reference solutions, multimodal data quality.

Teaching & Technical Communication: University-level instruction, mathematical exposition, technical writing.

AI & DATA EXPERIENCE

AI & Data Quality Evaluator | TELUS | 2026-present

- Evaluates speech and audio for naturalness, clarity, delivery style, and recording quality; applies task-specific guidelines and documents rating justifications.
- Assesses search results for relevance and usefulness against user intent; applies rating guidelines and writes comparative justifications.

Handshake | 2026-present

- Evaluates mathematical responses and proofs using structured rubrics for correctness, completeness, and logical consistency; documents errors and gaps in reasoning.
- Develops and reviews advanced mathematics prompts, assessment rubrics, and reference solutions, including problems in analysis and mathematical physics.
- Formalizes mathematical arguments in Lean 4 with Mathlib; checks project builds, theorem assumptions, and proof dependencies to support reproducible verification.

AI Data Annotator (Spanish - Mexico) | RWS | 2026-present

- Evaluates and tags Spanish (Mexico) data to improve AI assistant response quality.
- Reviews voice conversations for naturalness, turn-taking, interruptions, and audio quality; documents specific errors and assesses whether models correct them after feedback.
- Annotates and transcribes audio, video, images, and text; contributes recordings and written inputs and outputs for multimodal training datasets.

Robotics Data Collector - Quality Assurance | Remotasks | 2025-2026

- Collected robotics data and performed quality assurance for generative AI model projects.

AI Trainer | Outlier AI | 2024-2025

- Contributed to AI training projects requiring structured analytical and written work.

EDUCATION & PROFESSIONAL TRAINING

- 2008-2013: Bachelor's Degree in Mathematical Sciences, Faculty of Sciences, UNAM. Professional License: 14085315.
- 2014-2017: Master's Degree in Mathematical Sciences, Institute of Mathematics, UNAM. Professional License: 14030555.
- 2018-present: PhD in Mathematical Sciences (in progress), Institute of Mathematics, UNAM.
- 2023-present: Bachelor's Degree in Physics (in progress), Faculty of Sciences, UNAM.

PROFESSIONAL PROJECTS

The Mathematical Physics Project | Creator & Curator

- Created and maintains a growing digital library of self-contained notes, books, exercises, and reference materials for students, instructors, and researchers.
- Publishes and organizes downloadable resources across quantum theory, relativity, gauge theory, geometry, algebra, operator algebras, and classical mechanics.

TEACHING EXPERIENCE

- 2024: University Algebra I, Tiffin University (online).
- Differential and Integral Calculus, Southern New Hampshire University (online).
- Discrete Mathematics, Saint Leo University (online).
- 2023: Control Theory, General Relativity, Quantum Mechanics, Theoretical Mechanics, and Quantum Field Theory.
- 2019-2020: Riemannian Geometry, Master's Program in Mathematical Sciences.
- 2016-2017: Mathematics I, II, III, Instituto Nueva Inglaterra (secondary level).
- 2014-2016: Adjunct Professor, Linear Algebra I, Faculty of Sciences, UNAM.
- 2008-2009: Mathematics Instructor for Undergraduate Admissions, Faculty of Sciences, UNAM.

RESEARCH AFFILIATIONS

- Center of Mathematical Sciences and Applications (CMSA), Harvard University, Cambridge, Massachusetts.
- Max Planck Institute for Mathematics, Bonn, Germany.

SELECTED RESEARCH ACTIVITIES

- 2025: Participant, ICMAM Latin America Satellite Conference on Differential Geometry.
- Participant, Twinned Workshop on Quantum Field Theory and Topological Phases via Homotopy Theory and Operator Algebras, Max Planck Institute for Mathematics and CMSA.
- Presenter, Category Theory Seminar, Institute of Mathematics, UNAM.
- Member, PAPIIT Research Project: Structures in Tricategories: Monads and their Generalizations (IN110111).

LANGUAGES & TECHNICAL SKILLS

Languages: Spanish: Native | English: Advanced | French: Intermediate

Formal Verification: Lean 4, Mathlib, Lake; mathematical formalization, proof checking, and axiom dependency review.

Programming & Scientific Computing: Python, Java, MATLAB, Simulink, Anaconda.

Development Tools & Version Control: Visual Studio Code, Xcode, Git, GitHub, terminal/shell; project builds, script execution, and verification workflows.

Application & Web Development: Swift, SwiftUI, HTML, CSS, JavaScript, React, Vite; project experience with AI-assisted development.

Technical Writing & Publishing: LaTeX; preparation of mathematical documents and maintenance of a digital resource library with GitHub Pages.

AI-Assisted Workflows: Codex; assisted coding, debugging, mathematical formalization, and document preparation.